

# 2026 ISPE ANNUAL MEETING

August 29 -September 2, 2026

**MILAN, ITALY**

*Unlocking the Power of Pharmacoepidemiology  
to Improve Patient Health*

[2026ispe.eventscribe.net](https://2026ispe.eventscribe.net)



# Geospatial methods

Dr Chloé Corrie Hans Smit,  
*University of Technology, Sydney*

---



42nd Annual Meeting

Disclosures:  
none



## Contents

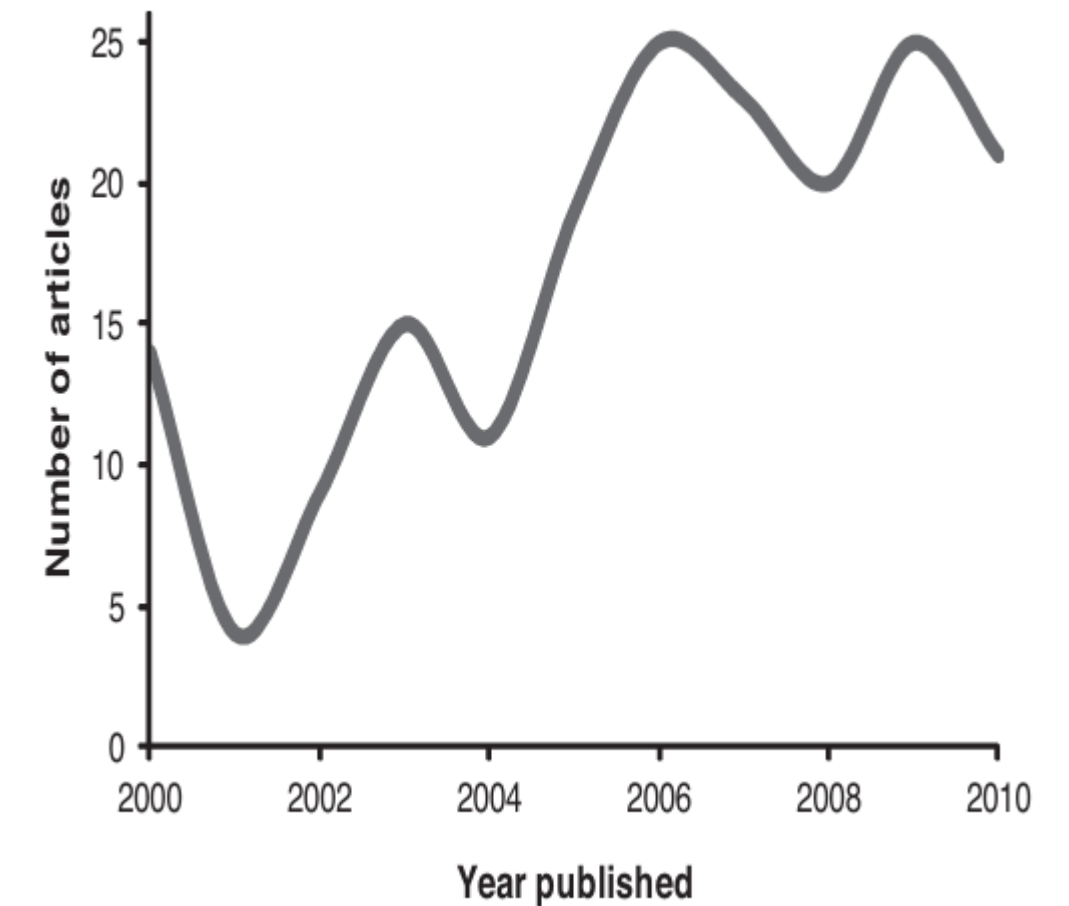
1. Introduction to spatial epidemiology
2. What is geospatial and geo-temporal analysis
3. Geospatial and geo-temporal examples
4. The geo-temporal analysis toolbox
5. Challenges of geo-temporal analysis
6. How to overcome these challenges

## Spatial Epidemiology

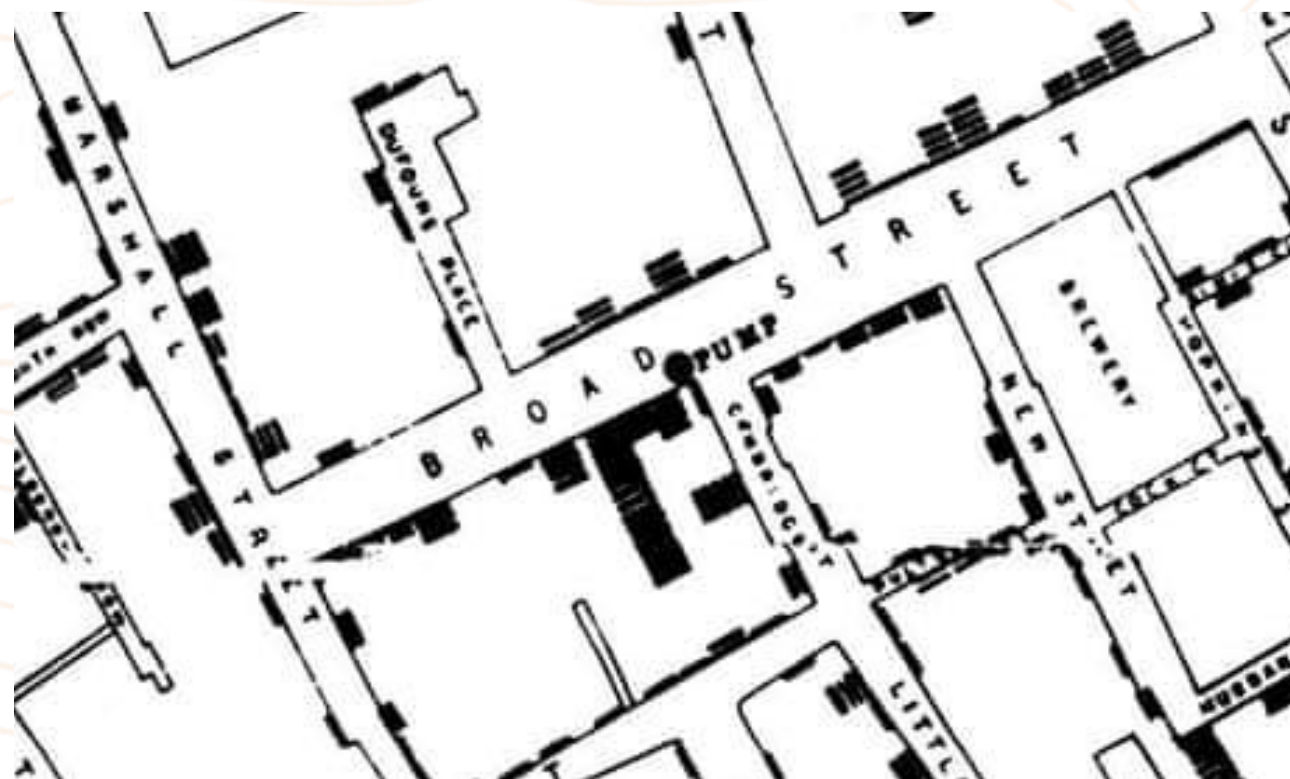
In subject area: Medicine and Dentistry

Spatial epidemiology is defined as a subcategory that quantifies and explains geographical variation in diseases, focusing on their association with socio-economic, genetic, demographic, behavioral, and environmental factors. It utilizes disease mapping to predict disease outcome patterns across various geographical areas, which can also help explore disease causation.

AI generated definition based on: [Clinical Trials and Tribulations, 2023](#)



Continuing to increase<sup>1</sup>



It is not a new field:  
John Snow and the  
Broad Street Cholera Map of 1854



## Definitions

Geospatial: investigates the geographical and **spatial spread** of data <sup>2</sup>

Geo-temporal: investigates the spread of geospatial data **over-time**

2. Fatima M, O'Keefe KJ, Wei W, Arshad S, Gruebner O. Geospatial Analysis of COVID-19: A Scoping Review. *International Journal of Environmental Research and Public Health*. 2021;18(5):2336.

## Reasons for doing geospatial analysis in pharmacoepidemiologic research

**Table 3**

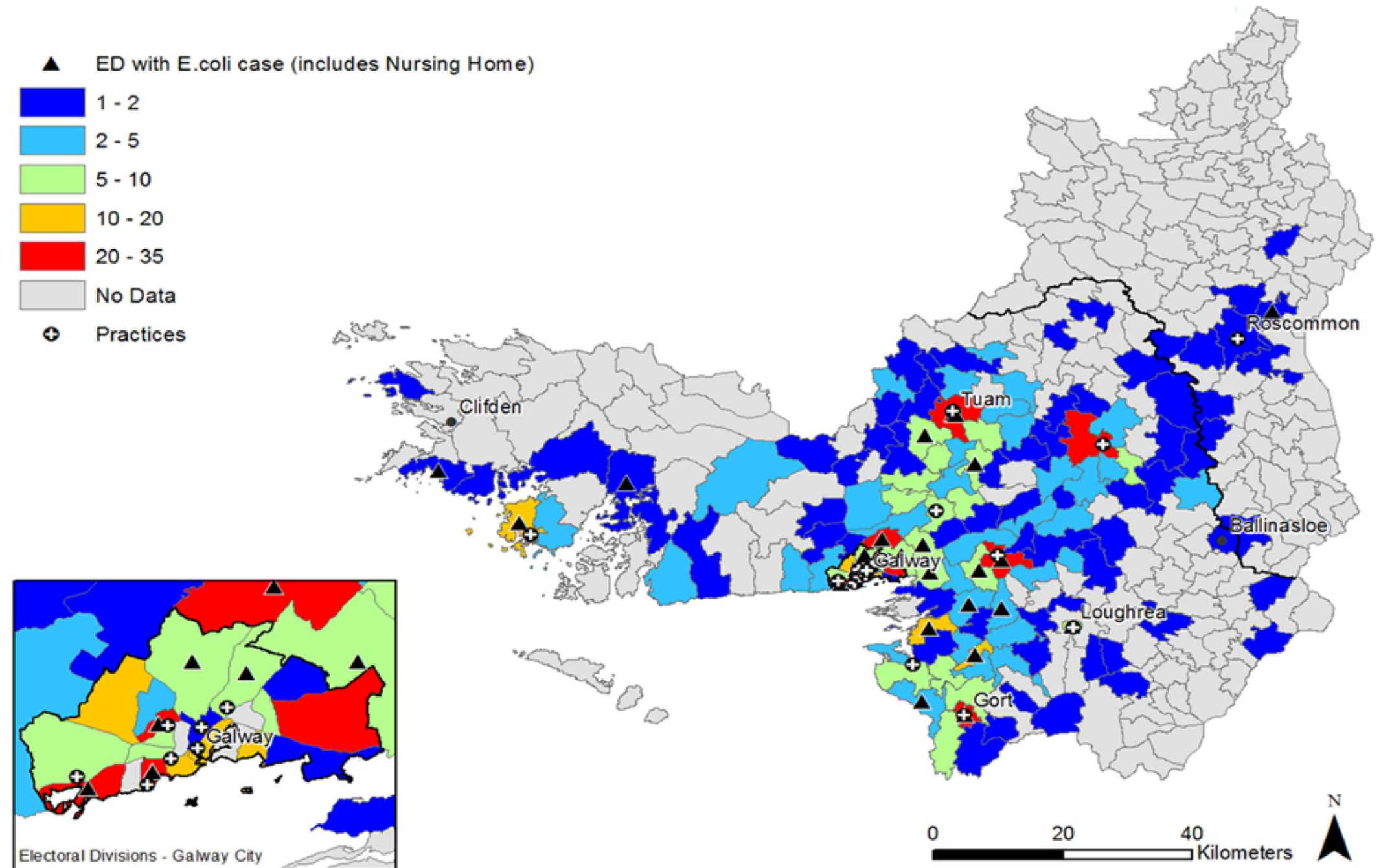
Author-reported motives and added value of the spatial or geographic approach used

| Added value                                                                            | n |
|----------------------------------------------------------------------------------------|---|
| Intervention and policy (n = 24)                                                       |   |
| Influence regional/local prescribing behavior (e.g. misdiagnosis, over/undertreatment) | 6 |
| Education                                                                              | 1 |
| Prevent or counteract drug abuse                                                       | 5 |
| Quality/efficiency of care                                                             | 5 |
| Access to care/medication                                                              | 5 |
| Give practitioners insight into the differences with their peers in other countries    | 1 |
| Insight into regional drug safety                                                      | 1 |
| Scale level (n = 6)                                                                    |   |
| Insight into suitability scale level                                                   | 1 |
| Data/prediction is more accurate at particular scale level                             | 5 |
| Explain regional variation/patterns (n = 7)                                            |   |
| Detect/explain local variation drug use/sales/health care use                          | 5 |
| Understand patterns of drug use/illness                                                | 2 |
| Explorative (n = 7)                                                                    |   |
| Explore an indication of regional differences/similarities                             | 6 |
| Find out whether findings from one study can be extrapolated to a larger region        | 1 |
| Other (n = 3)                                                                          |   |
| Regional data is used as surrogate for individual data                                 | 2 |
| Region as control variable                                                             | 1 |



## Example 1: Static mapping

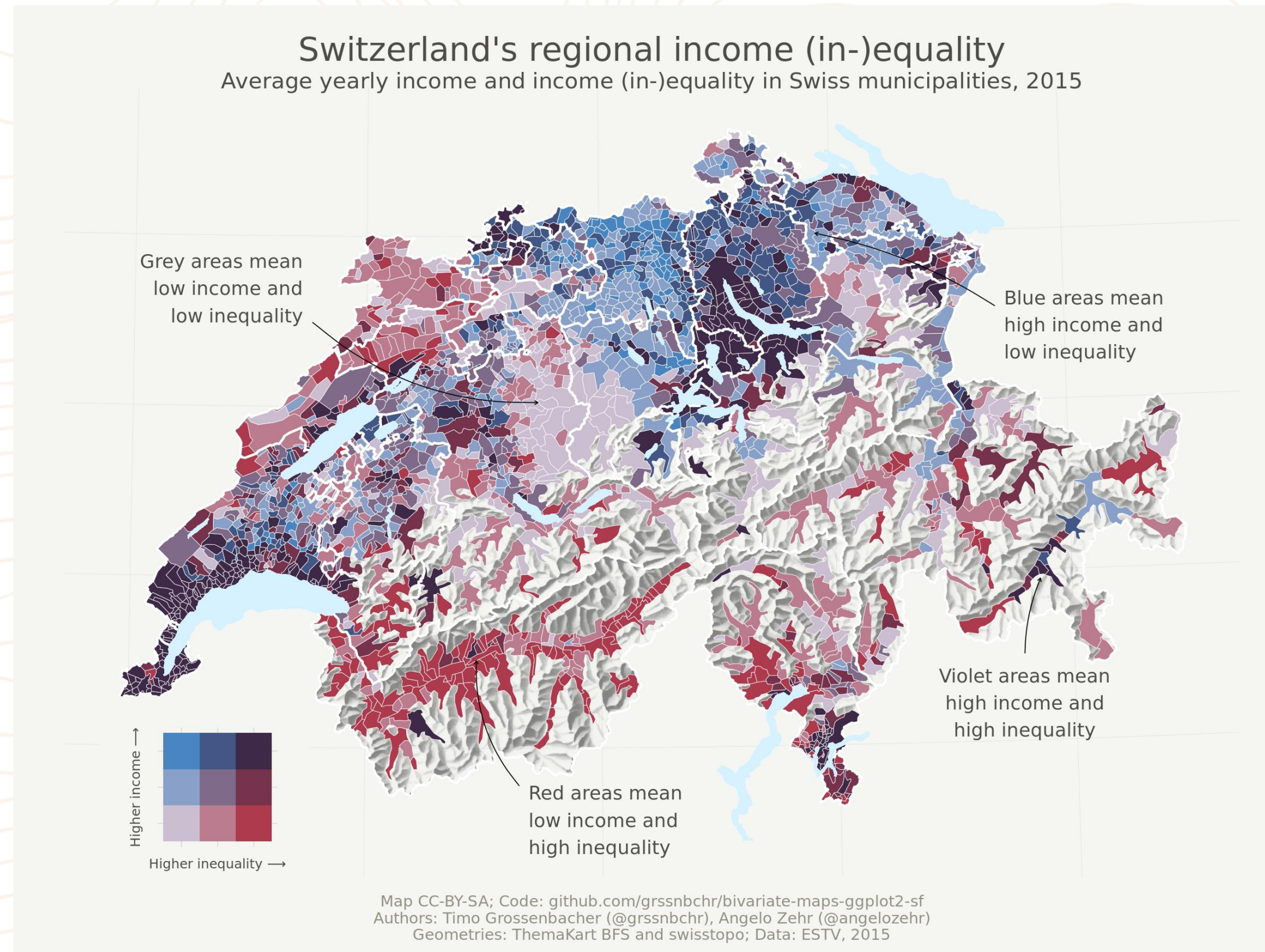
Figure 1. Distribution and frequency of *E. coli* urinary tract infection (UTI) cases, practices and nursing homes in the study region at electoral division level.<sup>4</sup>



4. Galvin, S.; Bergin, N.; Hennessy, R.; Hanahoe, B.; Murphy, A.W.; Cormican, M.; Vellinga, A. Exploratory Spatial Mapping of the Occurrence of Antimicrobial Resistance in *E. coli* in the Community. *Antibiotics* **2013**, *2*, 328-338. <https://doi.org/10.3390/antibiotics2030328>

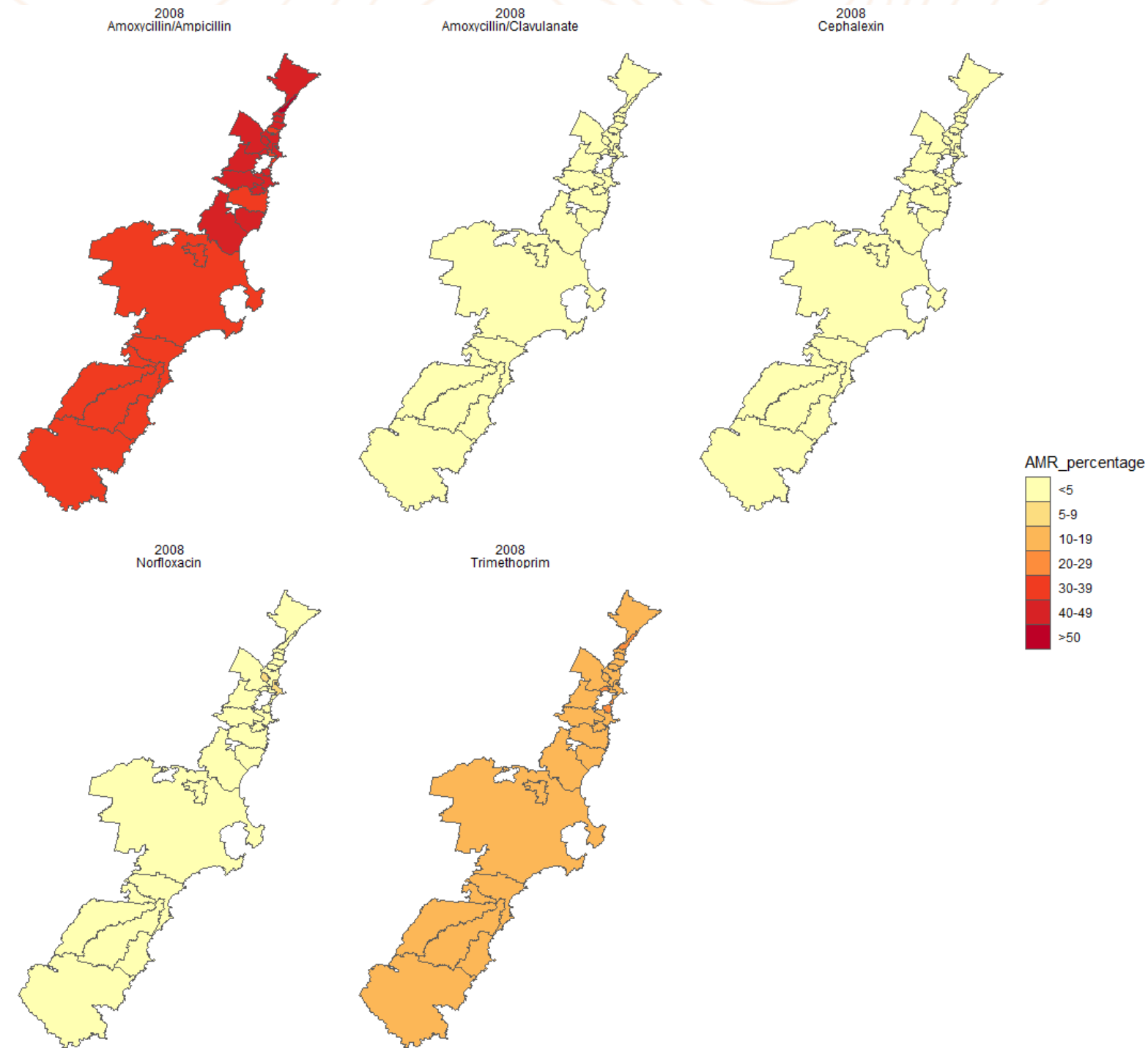


## Example 2: Bivariate maps





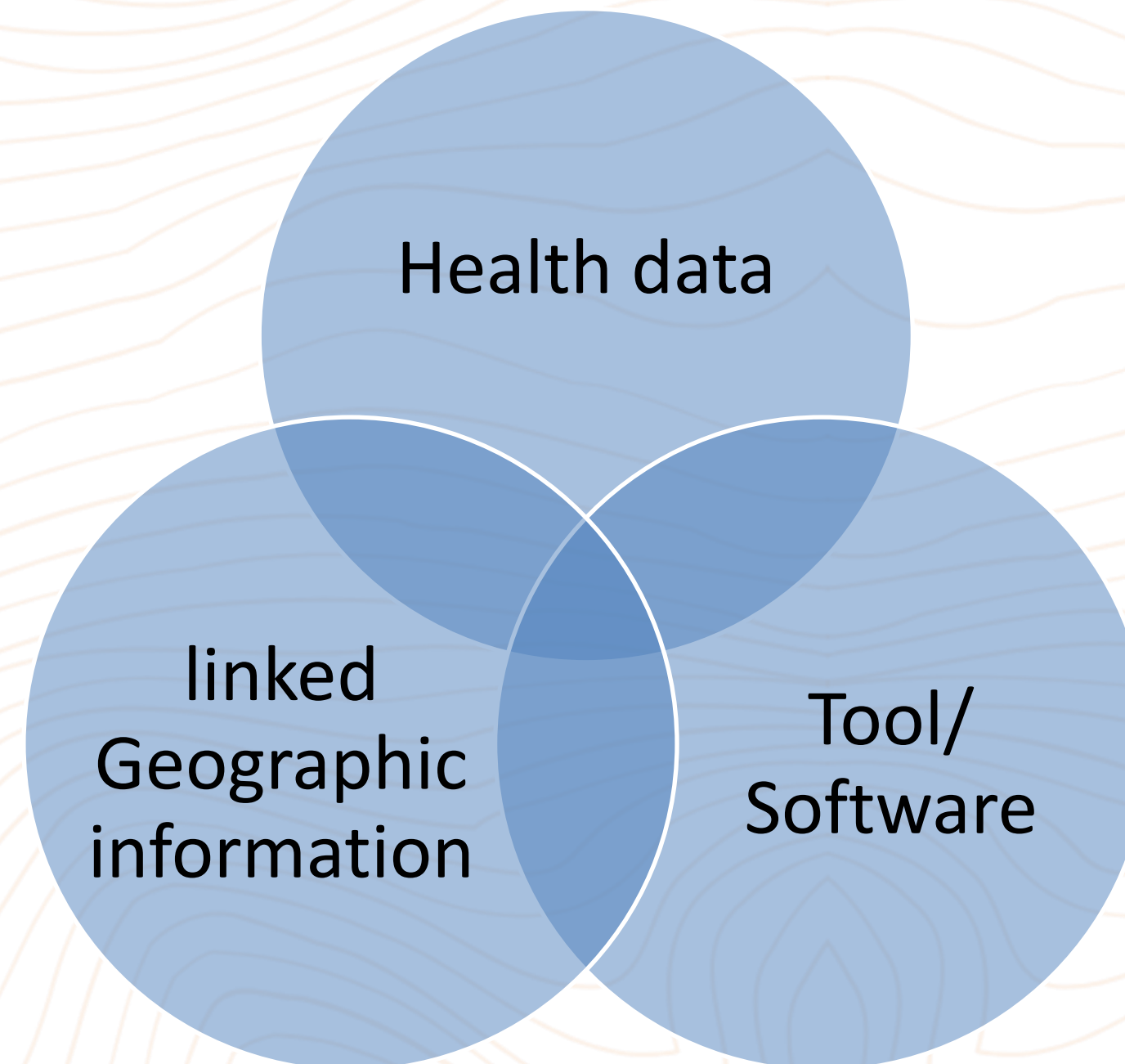
## Example 3: Adding time - geo-temporal variation





## The geo-temporal analysis toolbox

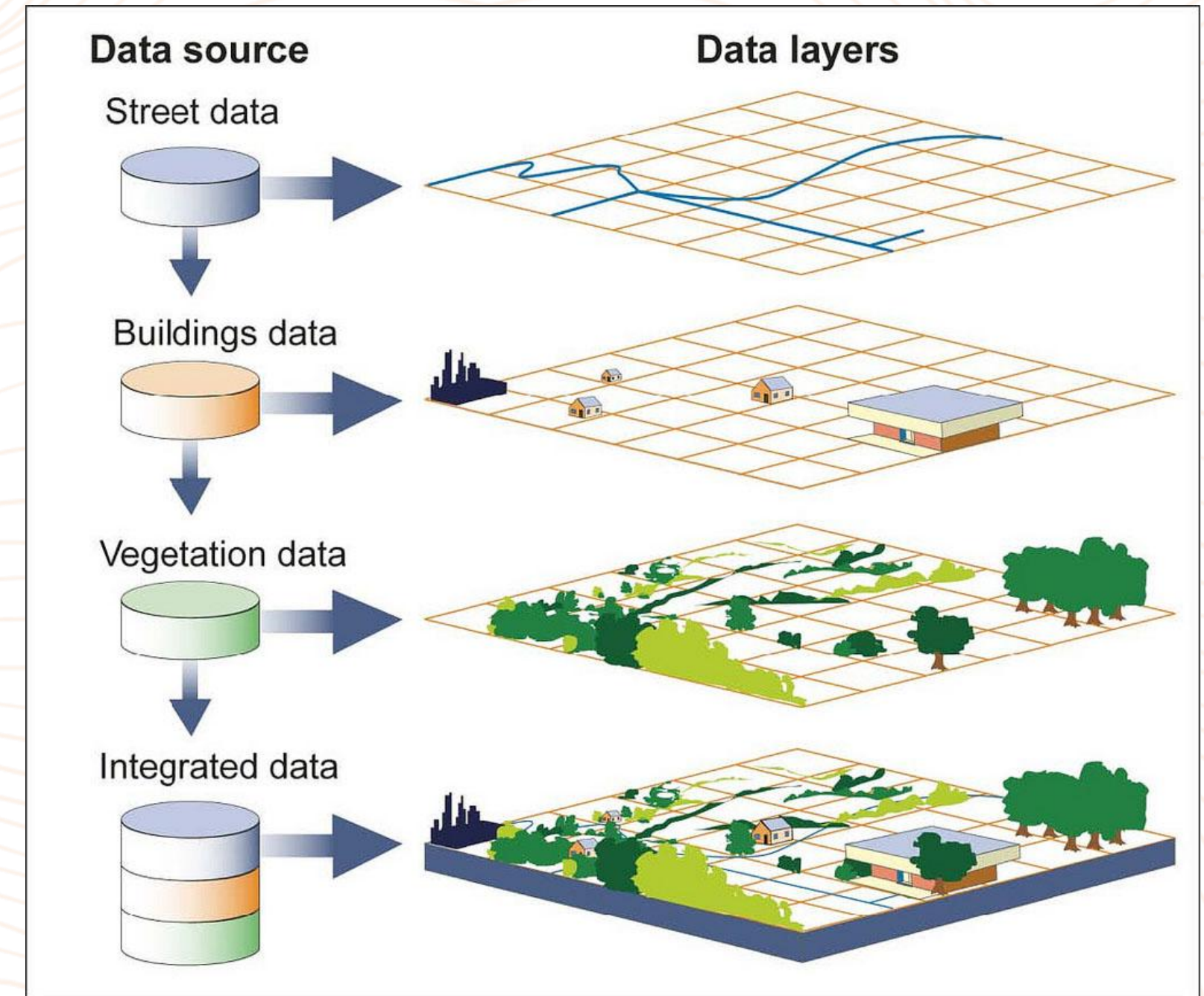
1. **What do you need?**
2. **How do you do it?**





## Geographic Information Systems <sup>7</sup>

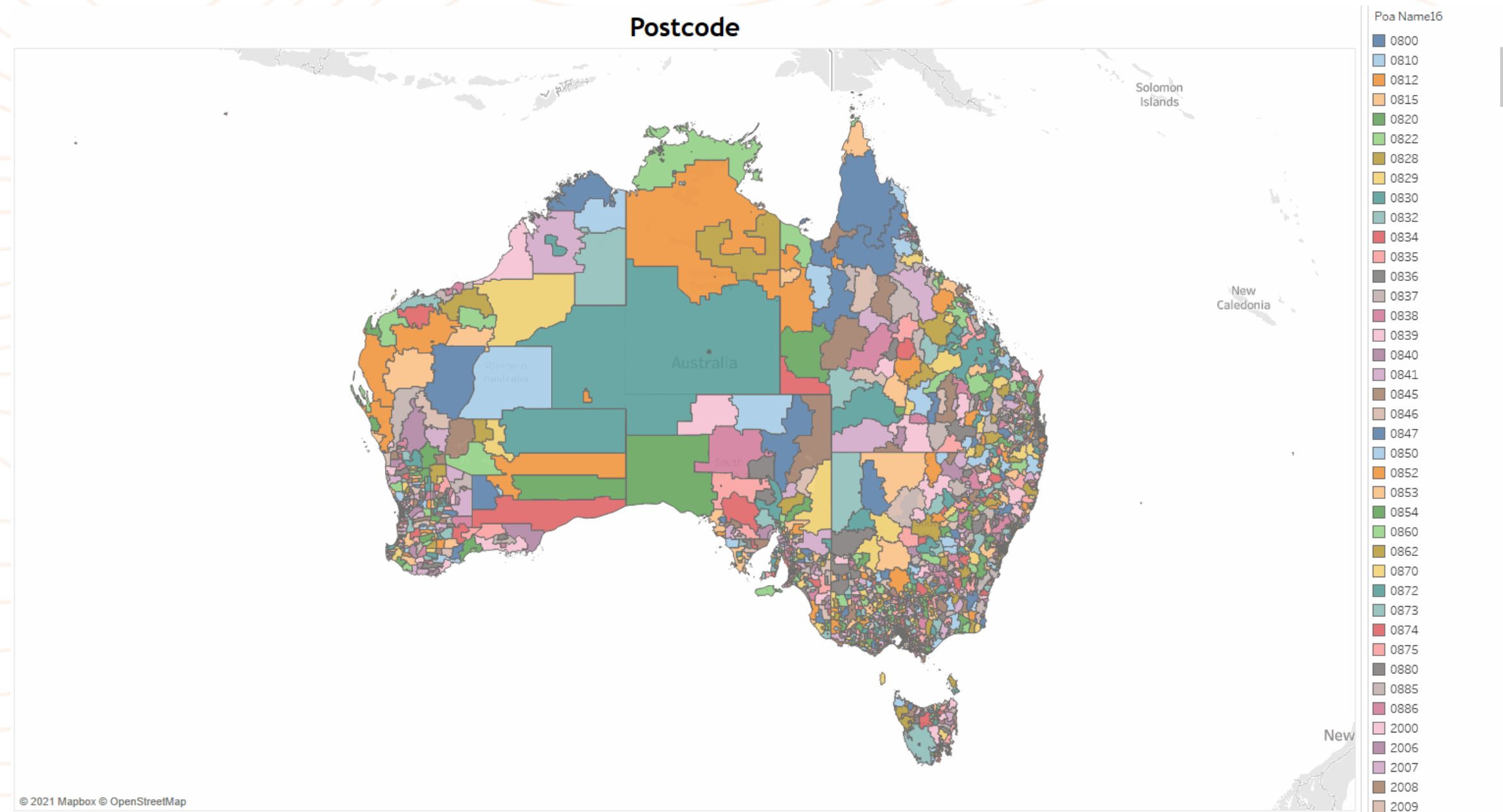
- Store, share, analyze, and visualize spatial data
- Integration of multiple layers of interdisciplinary spatial data
- Administrative boundaries and maps
- Examples: Google Earth, ArcGIS, Mapititude (Caliper )



Source: GAO.



## Geographic Information Systems: Administrative boundaries and maps <sup>8</sup>





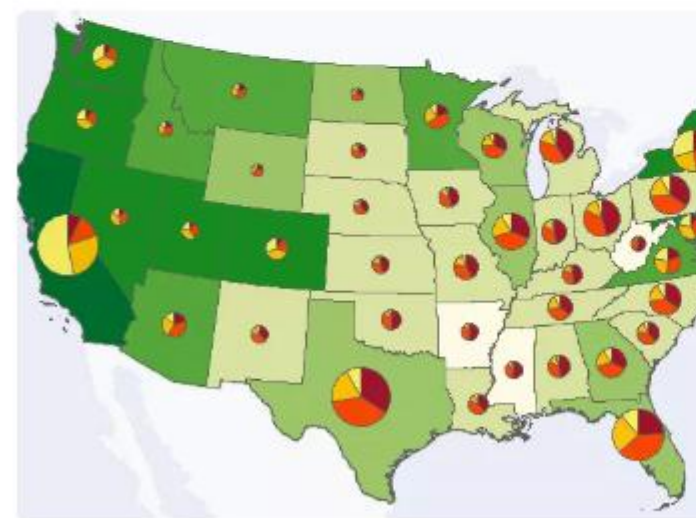
## Software examples:

### Maptitude Mapping Software

Desktop Online Map Your Data Analysis Data Examples API New Features Pricing Learn

#### Map Data

Maptitude includes extensive [geographic and demographic data](#) so you can get started as soon as you start using the application. Compare this to most mapping and GIS software that require separate purchases or the paying for credits to access geospatial information. Maptitude is integrated with the [best datasets](#) to support successful mapping and problem solving. Surprisingly few map tools provide th



Maptitude includes nationwide geography at can make den

Power BI Get started Work with data Create & share Admin & enterprise Developer Resources

Filter by title

- Create a new card visual with big number tiles
- Combo charts
- Doughnut charts
- Funnel charts
- KPI visuals
- Line charts
- Paginated report visuals
- Metrics visuals
- Maps
  - Shape map visuals (preview)
  - Filled maps (choropleth maps)
  - ArcGIS maps
  - Tips and tricks for map visualizations
- > Azure Maps in Power BI

### Create ArcGIS maps in Power BI

Article • 04/19/2023 • 18 contributors

#### In this article

- User consent
- Prerequisites
- Create an ArcGIS for Power BI map visualization
- Embed a map-enabled ArcGIS for Power BI report
- Show 7 more

APPLIES TO: Power BI Desktop Power BI service

ArcGIS for Power BI is a mapping visualization that helps you to enrich your data, reports, and mapping capabilities, demographic data, location analytics, and spatial analysis tools. Use the

### 3.2 Using ggplot2 to create maps

The easiest way to create spatial plots with ggplot is to use the `geom_sf()` function. By default there is no aesthetic mapping, but we can use arguments like `fill` to easily create choropleth maps.

```
usa_1 <- ggplot(data = honey2008) +
  geom_sf()
usa_2 <- ggplot(data = honey2008) +
  geom_sf(aes(fill = Price_per_lb)) +
  ggtitle(label = "The Honey crisis of 2008", subtitle = "Price per lb")
usa_1 / usa_2
```

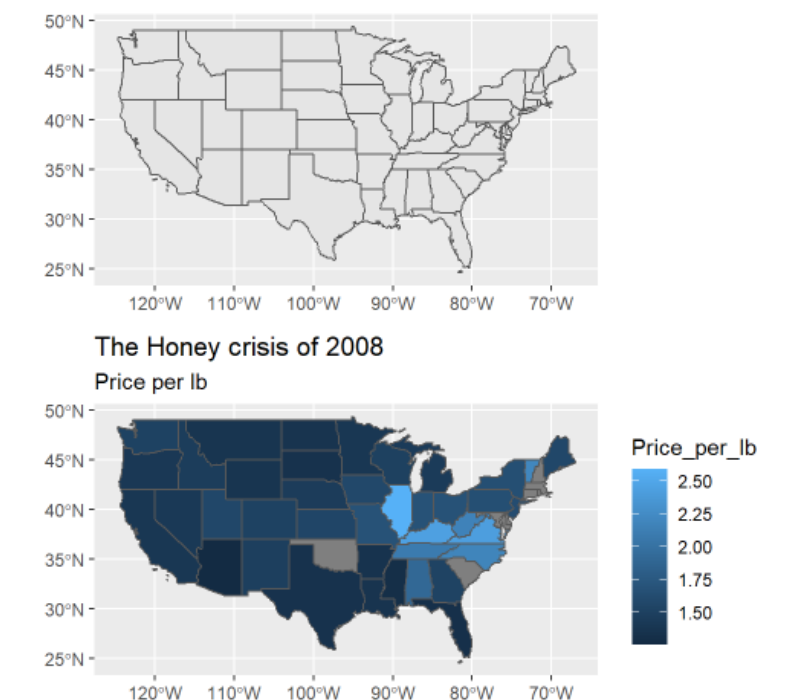


Figure 3.1: Basic use of ggplot2 for spatial data



## The geo-temporal analysis toolbox

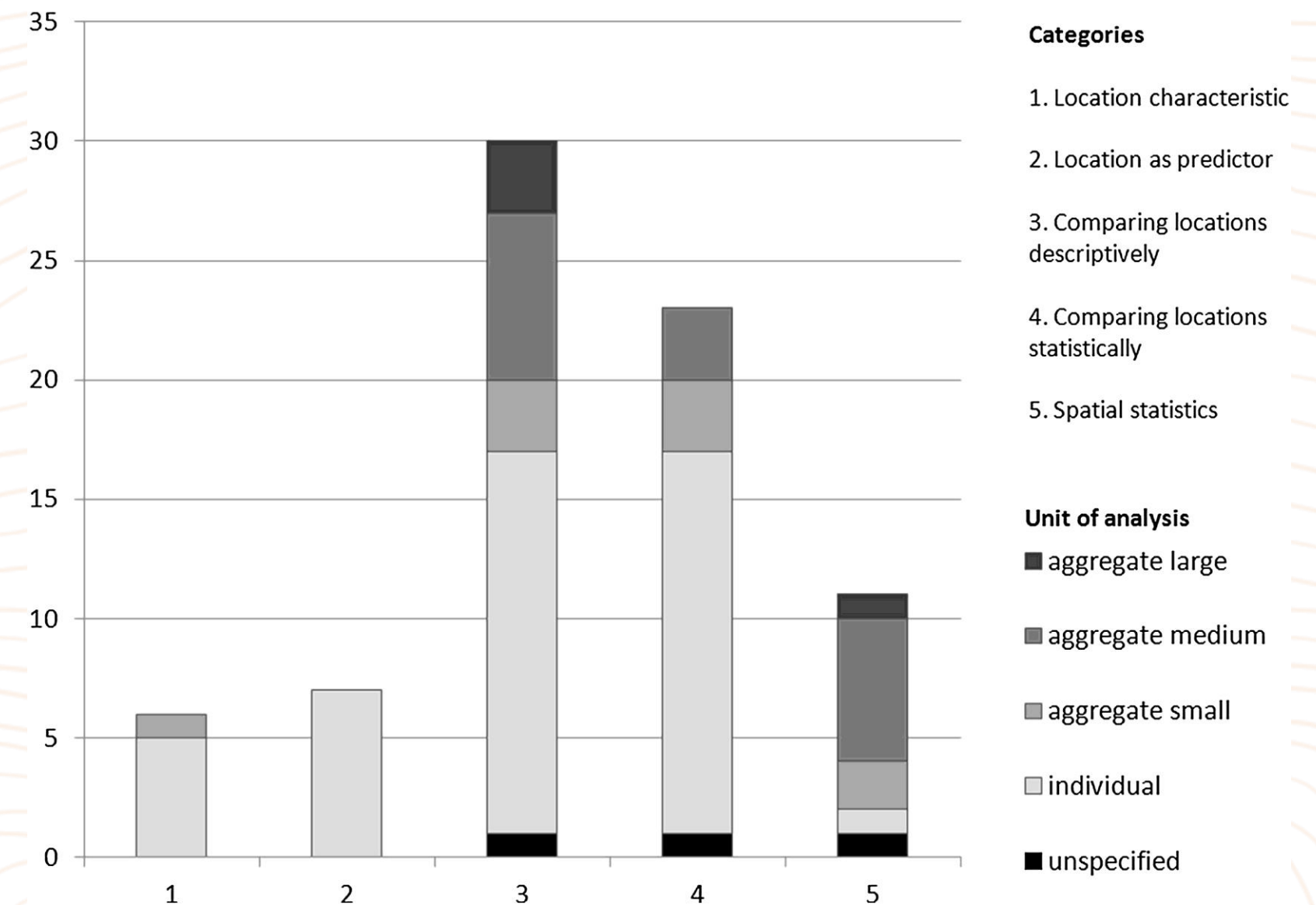
1. What do you need?
2. How do you do it?



## The geo-temporal analysis toolbox <sup>3</sup>

- Spatial proximity
- Aggregation methods
- Cluster detection techniques
- Spatial Interpolation and smoothing methods
- Multivariable spatial regression

Research methods and their unit of analysis





## Challenges when using geospatial analysis

1. Privacy
2. Big geography vs small population
3. Different geography levels across data
4. Ecological nature of analysis
5. Presentation of results



## 1. Privacy

**The problem:** location is identifying - a precise address plus a diagnosis can re-identify a patient, and the risk rises as the geographic unit shrinks.

- Analyse at the coarsest unit that still answers the question
- Apply minimum cell counts and suppress below them
- Aggregate any point (identifiable) data before publication



In Honolulu, a fire started by health authorities in an effort to "sanitize" the city's Chinatown in 1900 raged out of control, destroying much of the city. // Gabriel Bertram Bellinghausen via Wikimedia Commons

### The 'Chinese Flu' Is Part of a Long History of Racializing Disease

CARL ABBOTT MARCH 17, 2020

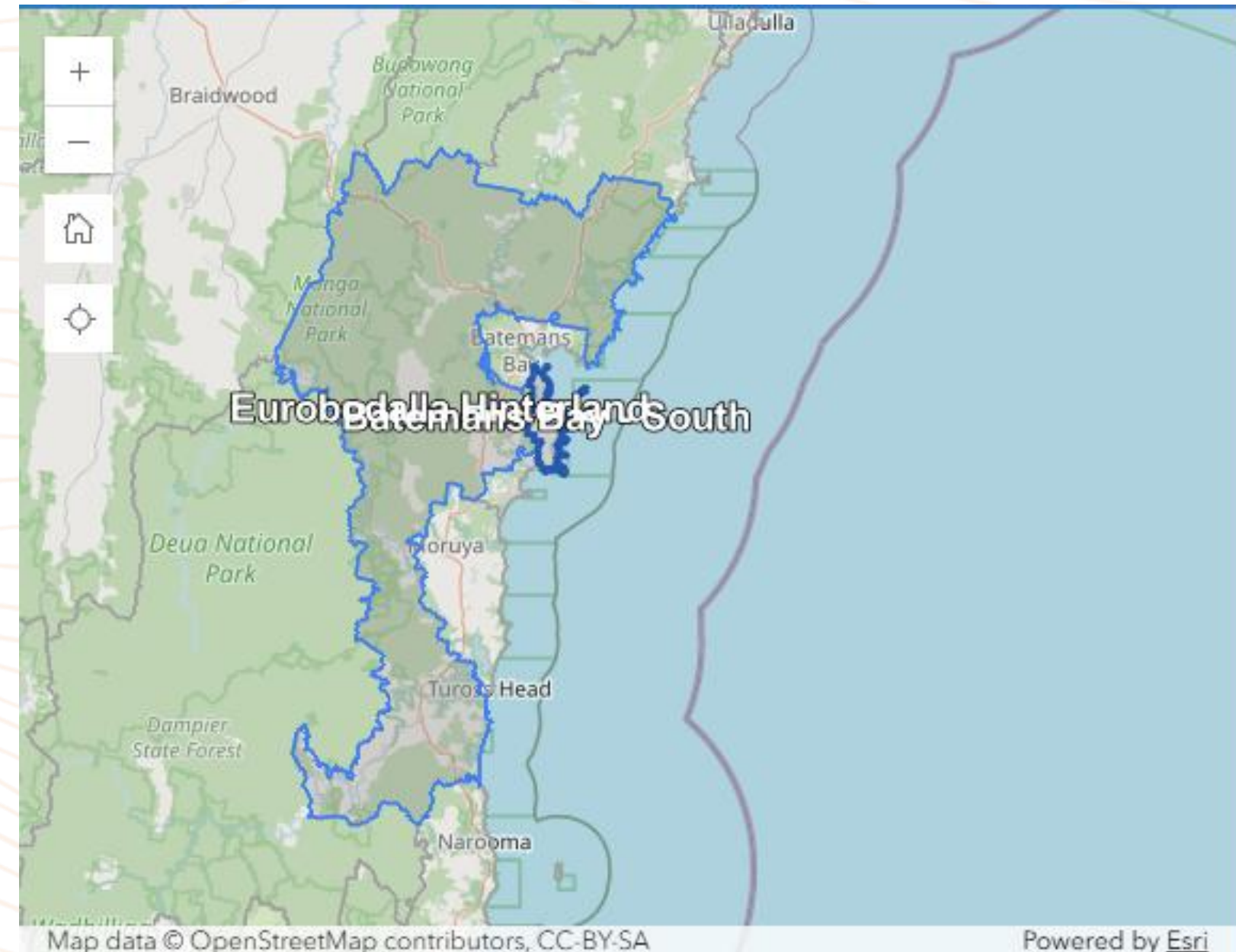
During a plague outbreak in 1899, officials in Honolulu quarantined and burned the city's Chinatown. Some Covid-19 talk today echoes their rhetoric.



## 2. Rural areas: Big geography, small population/ outcome

**The problem:** large polygons with few people - one extra case swings the rate, and empty land dominates the map visually.

- Use smoothing or Bayesian shrinkage to stabilise small-area estimates
- Report uncertainty alongside every estimate
- Consider population-weighted cartograms so visual area reflects people
- Suppress or flag areas below a minimum denominator

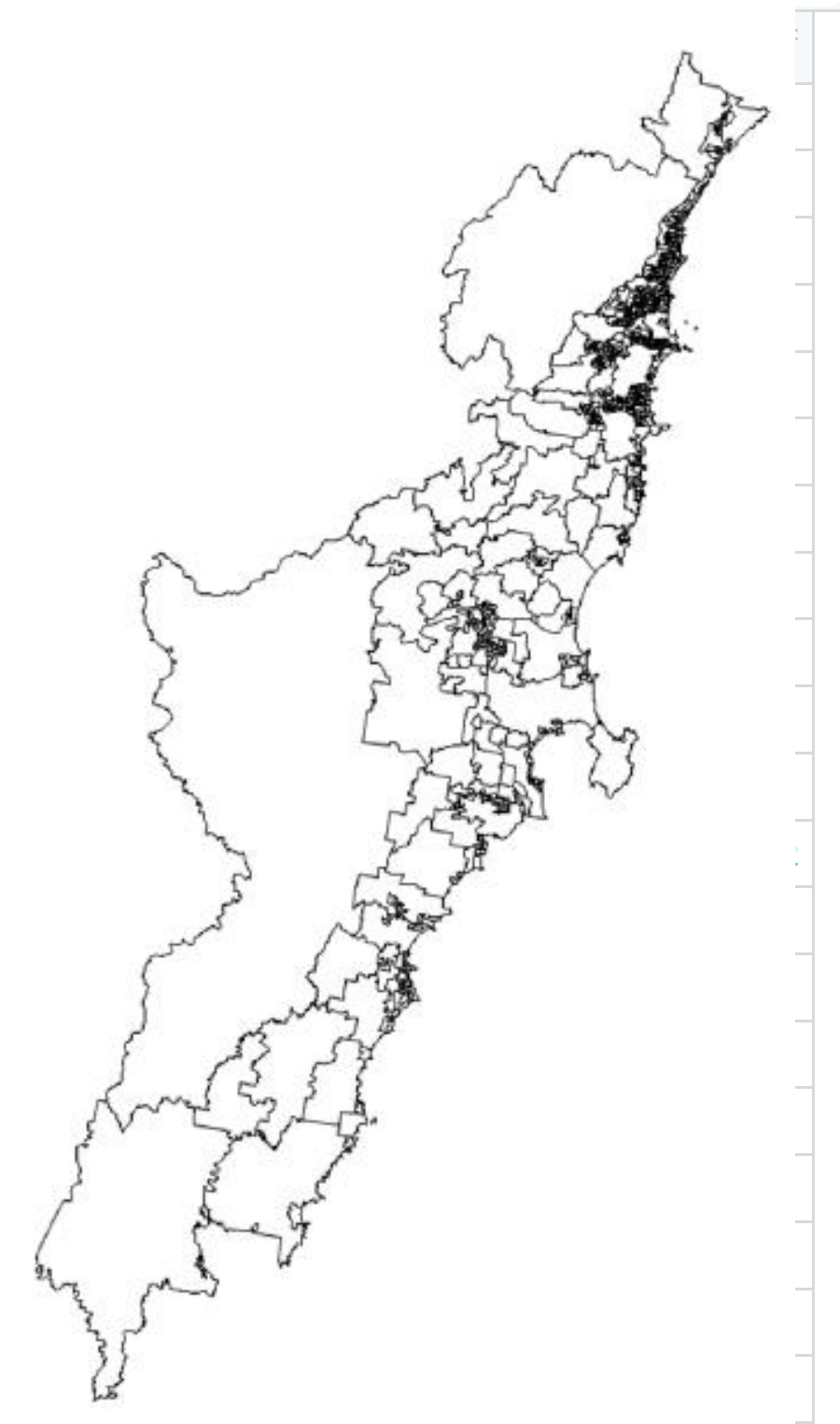




## 3. Geographical level differs across data

**The problem:** dispensing at postcode, covariates at statistical area, services at district - and none of them nest cleanly.

- Pick one analytic unit and justify it
- Convert between levels by population ratio, not by area
- Check the result holds at a second level





## 4. Ecological nature of analysis

**The problem:** an association between areas is not an association between individuals - the ecological fallacy.

- State the level of inference explicitly in every conclusion
- Use multilevel models to separate individual and area effects
- Treat area-level findings as hypothesis-generating unless the design supports more
- Watch for the modifiable areal unit problem - results shift with the unit



## 5. Presenting the results

**The problem:** maps are unusually persuasive, which makes them unusually easy to mislead with.

- Choose class breaks deliberately and state the classification used
- Keep colour scales consistent across panels being compared
- Show uncertainty rather than hiding it behind a single shade
- Always pair the map with the underlying numbers





Unlocking the Power of  
**Pharmacoepidemiology**  
to Improve Patient Health

 **ispe** International Society  
for Pharmacoepidemiology  
**42ND ANNUAL MEETING**

 **August 29 - September 2, 2026**  
 **Allianz Milan Convention Center**  
**Milan, Italy**



## Key take-aways

- Place is exposure data, not background noise
- Geospatial shows where; geo-temporal shows how it moves
- The tooling is mature - GIS, R and Python all get you there
- The real barriers are privacy, small numbers, mismatched geography and interpretation
- Each barrier has a practical mitigation - none of them is a reason not to start



**GET IN TOUCH**

Questions or collaboration:  
[chloe.smit@uts.edu.au](mailto:chloe.smit@uts.edu.au)